

JONATHAN S. HØJLEV

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in Jonathan Højlev

I am a passionate learner with a curiosity for all things natural sciences. In my studies I have found a passion for mathematics and theoretical computer science, lately focusing on graph theory and algorithmics. I have extensive experience in teaching and science communication and enjoy working in collaborative research environments.

EDUCATION

- 2024 — 2026 **M.Sc. in Computer Science and Engineering, Honour's Programme** DTU. GPA: 11.6/12
Focus: on Algorithms, Graph Theory, and Discrete Mathematics. I have followed the "Artificial Intelligence and Algorithms" specialization.
- 2/2025 — 6/2025 **Exchange in Vienna** TU Wien
I went to Vienna on exchange. It was a highly rewarding experience, where I got to do research in algorithmics with some of the very skilled people from TU Wien.
- 2021 — 2023 **B.Sc. in Software Engineering** DTU. GPA: 11.6/12
Focus: Algorithmics, Data Structures, and Software Development.
- 2019 — 2021 **Initial years at B.Sc. Quantitative Biology and Disease Modelling** DTU. GPA: 10.8/12
Focus: Cell biology and machine learning.

PUBLICATIONS

- Spring 2026 **Fair Correlation Clustering Meets Graph Parameters** LATIN 2026
Achieving FPT and XP results on the Fair Correlation Clustering problem, when parameterizing by graph parameters. Research initiated while on exchange in Vienna in spring 2025.

RELEVANT COURSES AND PROJECTS

- 2025 Autumn **Master's Thesis: Dynamic Graph Drawing** DTU
This project tries to review the techniques behind Level Planarity Testing of Level Graphs and to explore a dynamic approach to solving this problem.
- 2025 Spring **Graph Drawing Algorithms** TU Wien
This course teaches common graph drawing criteria and techniques for solving or optimizing the problem of creating a drawing for a graph belonging to common graph classes. The course simultaneously involved participating in a graph drawing contest [link] for GD25.
- 2025 Spring **Advanced Research in Algorithmics** TU Wien
Research project resulting in a submission to LATIN 2026.
- 2024 Autumn **Computational Geometry** DTU
This PhD course dives into computational geometry subjects such as convex hulls, triangulations, voronoi diagrams, Frechet distance and more.
- 2024 Autumn **Graph Theory 2** DTU
The course includes a number of classical results such as the theorems of Tutte, Ramsey, Turan, Kuratowski, Brooks, Dirac, Smith, and Vizing, and also the Jordan Curve Theorem. The course also includes more recent themes, for example list-colorings.
- 2024 Autumn **Computationally Hard Problems** DTU
Studying algorithmic problem classes such as NP, performing problem reductions from known NP-Hard problems, and analyzing randomized algorithms.
- 2024 Spring **Function Spaces and Mathematical Analysis** DTU
Understanding and applying advanced mathematical concepts regarding function spaces, norms, convergence, L^p spaces, Hilbert spaces, Fourier transforms, and more.
- 2023 Autumn **Algorithmic Techniques for Modern Data Models** DTU
To know, apply, analyze, and design algorithms in modern data models, such as the streaming model, distributed algorithms, and sketches such as distance oracles.

EXPERIENCE

2020–2024 part time	Teaching Assistant, Advanced Engineering Mathematics 1 I have worked as a TA in Mathematics 1 at DTU for most of my studies (7 semesters). This included supervising and examining a large mathematics project twice. For this work, I got nominated by the students as "Teaching Assistant of the year 2021".	DTU Compute
2020–2024 twice a year	Organizer of Mathematics Walkthrough Lecture I have personally arranged and held a biannual walkthrough lecture of the Mathematics 1 course for around 1000 students—something I am very proud of. Recordings here: Link .	DTU
9/2024–12/2024 part time	Teaching Assistant, Algorithms and Data Structures 2 Teaching assistant in the course algorithms and Data Structures 2	DTU Compute
9/2024–12/2024 part time	Teaching Assistant, Computer Programming Teaching assistant in the introductory Python programming course at DTU.	DTU Compute
2/2023–5/2023 part time	Teaching Assistant, Introductory Programming Teaching assistant in an introductory programming course in Java.	DTU Compute
2/2022–5/2022 part time	Teaching Assistant, Algorithms and Data Structures 1 Teaching assistant in the course algorithms and Data Structures 1	DTU Compute
12/2018–12/2021 part time	Science Communicator At the science and technology museum, Experimentarium, I worked with guest service and as a science communicator to audiences in all ages. This taught me presentation and teaching skills, as well as first aid and crisis management.	Experimentarium

LANGUAGES

Danish — native
English — fluent
French — basic understanding

HOBBIES

I love playing around and learning new things. My hobbies include sailing, rope climbing, choir singing, and the collaborative storytelling game of Dungeons & Dragons.

I love meeting new people, exploring other cultures and languages, and learning new fun facts about the world.

NON-PROFIT

As a volunteer in our student union PF at DTU, I have worked with planning the Study Start for approx. 2500 new students by managing and coordinating 300 volunteers in collaborating with a team of other coordinators. For my work on this as well as in areas like our student council and as voluntary TA, I received the "Initiator of the Year" award in 2022.